

12.848

EE25BTECH11002 - Achat Parth Kalpesh

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Question

The tangent plane to the surface $x^2 + y^2 + z = 9$ at the point $(1, 2, 4)$ is

- ① $2x + 4y + z = 14$
- ② $4x + 2y + z = 12$
- ③ $x + 4y + 2z = 17$
- ④ $4x + y + 2z = 14$

Theoretical Solution

The equation of the surface is

$$\mathbf{x}^T \mathbf{V} \mathbf{x} + 2\mathbf{u}^T \mathbf{x} + f = 0 \quad (1)$$

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}, \quad f = -9 \quad (2)$$

The tangent plane is given as

$$(\mathbf{V}\mathbf{q} + \mathbf{u})^T \mathbf{x} + \mathbf{u}^T \mathbf{q} + f = 0 \quad (3)$$

Theoretical Solution

where

$$\mathbf{q} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \quad (4)$$

$$\left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix} \right)^{\top} \mathbf{x} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \end{pmatrix}^{\top} \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = 9 \quad (5)$$

$$\begin{pmatrix} 1 \\ 2 \\ \frac{1}{2} \end{pmatrix}^{\top} \mathbf{x} = 7 \quad (6)$$

$$\implies 2x + 4y + z = 14 \quad (7)$$

```
#include <stdio.h>
#include <math.h>

// Function to compute Z values for the surface:  $z = 9 - x^2 - y^2$ 
void compute_surface(int n, double *x, double *y, double **Z) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            Z[i][j] = 9.0 - x[i]*x[i] - y[j]*y[j];
        }
    }
}
```

```
// Function to compute Z values for the tangent plane:  $z = 14 - 2x - 4y$ 
void compute_plane(int n, double *x, double *y, double **Z) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            Z[i][j] = 14.0 - 2.0*x[i] - 4.0*y[j];
        }
    }
}
```

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
import ctypes

# Load the shared library
lib = ctypes.CDLL('./formula.so')

# Array size
n = 100
x = np.linspace(-3, 3, n)
y = np.linspace(-3, 3, n)

# Create empty Z arrays
Z_surface = np.zeros((n, n), dtype=np.float64)
Z_plane = np.zeros((n, n), dtype=np.float64)
```

Convert to C pointers

```
X_ptr = x.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
Y_ptr = y.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
Z_surface_ptr = Z_surface.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
Z_plane_ptr = Z_plane.ctypes.data_as(ctypes.POINTER(ctypes.c_double))
```

Define the functions argument types

```
lib.compute_surface.argtypes = [ctypes.c_int,
                                ctypes.POINTER(ctypes.c_double),
                                ctypes.POINTER(ctypes.c_double),
                                ctypes.POINTER(ctypes.POINTER(ctypes.c_double))]

lib.compute_plane.argtypes = [ctypes.c_int,
                              ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.c_double),
                              ctypes.POINTER(ctypes.POINTER(ctypes.c_double))]
```


Python Code

```
# Allocate row pointers for C 2D array
Z_surface_c = (ctypes.POINTER(ctypes.c_double) * n)()
Z_plane_c = (ctypes.POINTER(ctypes.c_double) * n)()
for i in range(n):
    Z_surface_c[i] = Z_surface[i].ctypes.data_as(ctypes.POINTER(
        ctypes.c_double))
    Z_plane_c[i] = Z_plane[i].ctypes.data_as(ctypes.POINTER(
        ctypes.c_double))

# Call C functions
lib.compute_surface(n, X_ptr, Y_ptr, Z_surface_c)
lib.compute_plane(n, X_ptr, Y_ptr, Z_plane_c)

# Point of contact
px, py, pz = 1, 2, 4

# Plot
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')
```

Python Code

```
X, Y = np.meshgrid(x, y)
ax.plot_surface(X, Y, Z_surface, alpha=0.5)
ax.plot_surface(X, Y, Z_plane, alpha=0.5)
ax.scatter(px, py, pz, s=60)

ax.text2D(0.85, 0.90,
          "Surface (blue):  $x^2 + y^2 + z = 9$ \n",
          "Tangent Plane (orange):  $2x + 4y + z = 14$ ",
          transform=ax.transAxes,
          fontsize=10,
          bbox=dict(boxstyle="round", alpha=0.5))
ax.text(px, py, pz + 1, "Point (1,2,4)", fontsize=10)
ax.set_xlabel('X - axis')
ax.set_ylabel('Y - axis')
ax.set_zlabel('Z - axis')
ax.set_title('Surface  $x^2 + y^2 + z = 9$  and Tangent Plane  $2x + 4y + z = 14$ ')
plt.show()
```

Surface $x^2 + y^2 + z = 9$ and Tangent Plane $2x + 4y + z = 14$

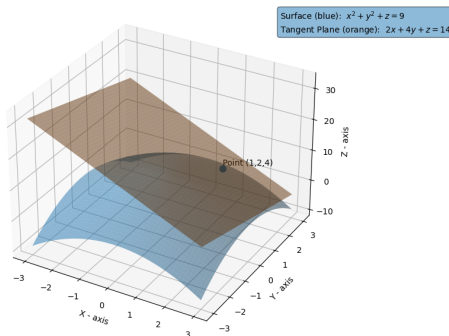


Figure: Graph